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REVIEW ARTICLE

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Factors that affect health information technology adoption by seniors: A systematic review

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Abstract

The number of seniors and prevalence of chronic conditions are increasing worldwide, resulting in more pressure on health systems. Health Information Technologies (HIT) present opportunities to support the healthcare needs of seniors. Although prior studies have investigated HIT and seniors, it remains unclear what factors significantly affect the adoption of different HIT by elderly people in the community. A Systematic Review (SR) was conducted between December 2017 and February 2018 (with a search update in 2018–2019) to critically appraise and synthesise existing evidence on HIT adoption factors among seniors. Following the PRISMA guidelines, five major databases were consulted (PubMed, Medline, CINAHL, Scopus and Web of Science). The inclusion criteria consisted of empirical studies, published in English, and reporting impacts of specific factors on HIT adoption among seniors in the community. A total of 41 studies were included in this review, mostly published between 2014 and 2017 in Europe and the US; the level of evidence in these studies was low to moderate. The factors that affect HIT adoption did not differ across types of technologies or age groups. The findings reveal that seniors adopt HIT that are *perceived as useful* and requiring *low effort* commitment; *price/cost* value were reported as adoption barriers. *Social influence*, *facilitating conditions*, *senior-friendly product design*, *self-efficacy*, *Intrinsic Technology Quality*, *experience/training* and *technology anxiety* may affect HIT adoption by seniors, although the evidence on these impacts remains weak and limited. Mixed and inconclusive evidence was observed on the impacts of *socio-demographic variables*, *health condition*, *habit* and *privacy/security*. Given the reported low level of HIT adoption among seniors, we call for more rigorous research in this area using a 'senior-centred' approach, which takes into account the discourse/interaction between seniors and their collective environment to better understand the factors that affect their technology adoption and address their needs.

KEYWORDS

community care, health information technology, PRISMA guidelines, seniors/elderly, systematic review, technology adoption

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1 | INTRODUCTION

Population ageing is a worldwide phenomenon resulting in demands at the social and health care levels (Breyer, Costa-Font, & Felder, 2010). For example, Canada's healthcare system is faced with challenges associated with population ageing and an increasing number of elderly people (also referred to as seniors in this review), with 75%–80% of them living with one or more chronic health conditions (Canadian Medical Association, 2016). Other countries (e.g. Japan, Italy) have also observed a considerable growth in their elderly populations, leading to increased demands on their healthcare systems (OECD, 2018).

In the light of these changes, there is an increasing need to shift healthcare services to the community while ensuring adequate support for them. Prior studies discussed the seniors' preference for living independently in their own homes or communities, as long as they get the necessary support and services to address their health needs (e.g. Claes, Devriendt, Tournoy, & Milisen, 2015; Satariano, Scharlach, & Lindeman, 2014; Tran, Buckley, Bertera, & Gonzales, 2009). Health Information Technologies (HIT) present opportunities to address the healthcare needs of seniors in the community and support them through innovative technologies (Alsulami & Atkins, 2016; Jaana, Tamim, Sherrard, & Pare, 2017; Vassli & Farshchian, 2017). These technologies consist of a variety of applications and tools (e.g. Information Communication Technology (ICT), telehealth, Electronic Health (E-Health)) that may address the challenges that elderly people face when living with chronic conditions and different levels of frailty (Choi et al., 2014; Hamblin, 2017; Keränen et al., 2017; Zhang et al., 2014).

A number of reviews on HIT and seniors have been published in the literature, which mostly focused on HIT effectiveness in assisting seniors (e.g. Botsis & Hartvigsen, 2008; Khosravi & Ghapanchi, 2016; Loader, Hardey, & Keeble, 2008) or examined seniors' acceptance of ageing-in-place technologies (e.g. Fischer, David, Crotty, Dierks, & Safran, 2014; Peek et al., 2014). To our best knowledge, no prior review has specifically examined the different adoption models in the literature in relation to various types of technologies, nor synthesised the evidence on the factors that significantly affect the adoption of these technologies by seniors. This review addresses this gap and complements previous systematic reviews in this area by critically appraising and synthesising the evidence on the factors that affect the adoption of different types of technologies by seniors in the community. Specifically, it addresses the following questions:

- What are the research models that are considered in the literature in relation to HIT adoption and seniors, and do these models vary across various types of technologies studied?
- What is the empirical evidence on the factors that affect HIT adoption by seniors?
- Do the factors affecting the adoption of HIT by seniors vary across types of technology?

What is known about the topic

- There is a growing number of emerging HIT that can support seniors' health care needs in the community.
- There is an overall low level of HIT adoption by seniors in the community, with limited knowledge about the major adoption factors for different HIT types.

What this paper adds:

- There is evidence of significant positive impacts of high *performance expectancy*, low *effort expectancy* and low *price/cost value* on HIT adoption by seniors. The findings are inconsistent and inconclusive for other factors given the low to moderte level of evidence.
- HIT adoption research should use a 'senior-centred' approach, which takes into account the discourse/interaction between elderly people and their collective environment.

2 | METHODOLOGY

Following the PRISMA guidelines (Moher, Liberati, Tetzlaff, & Altman, 2009), we applied the recommended step-wise approach to conduct this SR. Five major scientific databases (Web of Science, Scopus, Medline, Pubmed and CINAHL) were searched between December 2017 and February 2018; an additional search update was further conducted until December 2019. The following combination of keywords was used: (a) adoption (e.g. usability, use); (b) elderly (e.g. seniors, ageing); (c) health information technology (e.g. E-Health, ICT); and (d) living in the community (e.g. home, retirement home; see Appendix 1 for details on used keywords and their combinations). We followed the PICO (Population, Intervention, Comparison and Outcomes) model to frame the research questions as follows: (a) the population consisted of seniors living in the community; (b) the intervention was HIT (e.g. ICT, telemonitoring, E-Health); (c) the comparison was the absence of HIT and (d) the outcome was reported HIT adoption and intention to use by seniors.

The inclusion criteria consisted of: (a) empirical studies published in English, (b) studies that considered seniors in the community, and (c) studies that reported results on adoption factors that affect HIT adoption by elderly people. Examples of adoption factors included, among others, performance expectancy, effort expectancy, social influence etc. Exclusion criteria included non-empirical articles (commentaries, editorials, workshop summaries, expert opinion articles) and articles published in other languages. Overall, 1,573 articles were selected from all five databases (Figure 1). After removing duplicates and performing title/abstract screening, the total number of studies was reduced by 631 and 893 articles respectively. The content review further excluded 20 articles. Following the snowballing method and the search update, a

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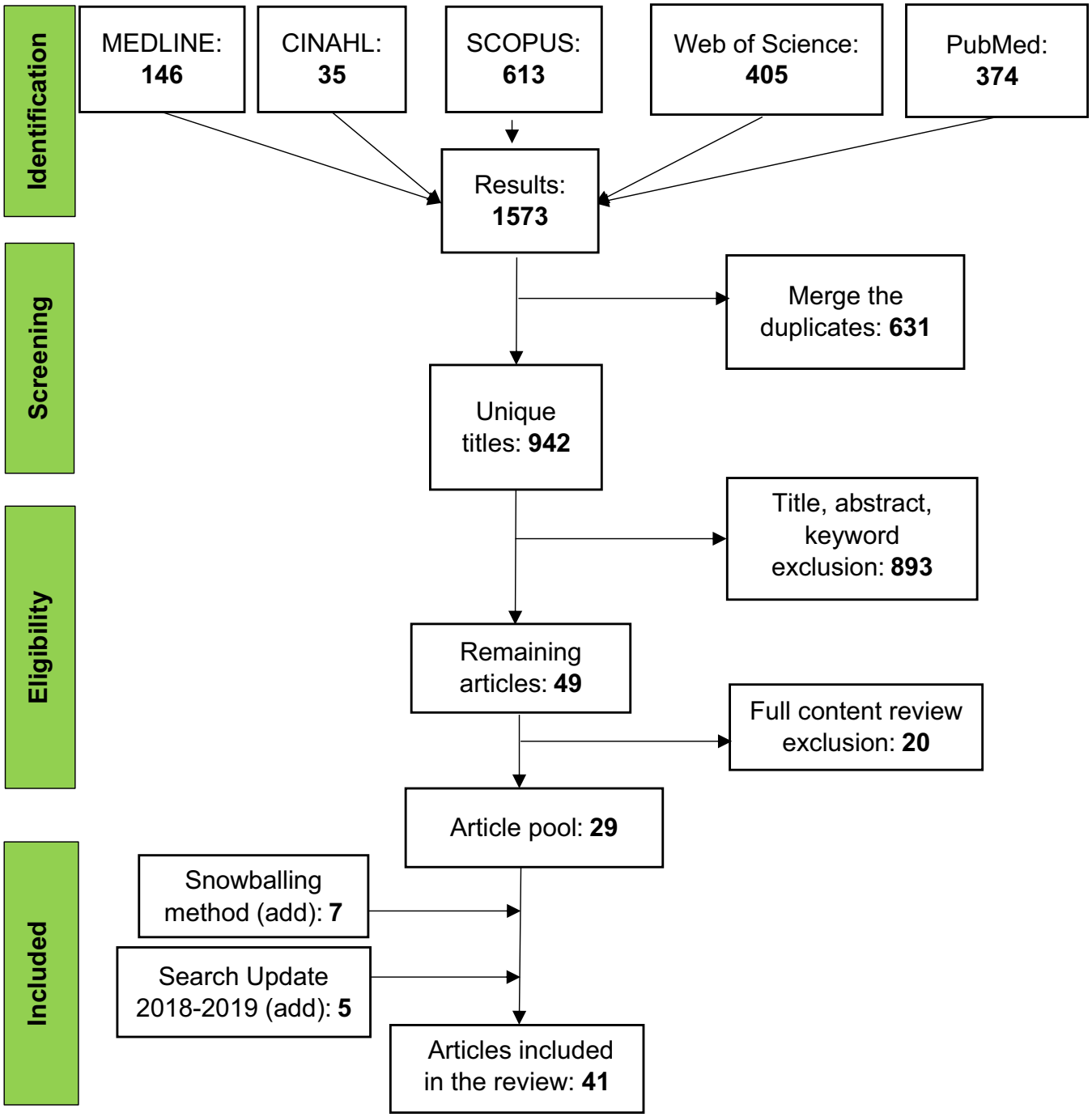

FIGURE 1 Flowchart representing the search and review steps (PRISMA guidelines, Moher et al., 2009) [Colour figure can be viewed at wileyonlinelibrary.com]

total of 41 articles were included in this review (Figure 1). The relevant information from these articles was extracted using a coding scheme that covered the following: (a) general information (journal information, study type); (b) methods and technology (data collection techniques, level of evidence, sample size, technology-orientation); and (c) results related to adoption (adoption factors, significant impacts).

To assess the quality of the studies, we adapted the method proposed by Thomsen and Greer to evaluate and rank the level of evidence in the selected articles (Thomsen & Greer, 2015). The studies were ranked on five levels (in line with the hierarchy of 'evidence-based practice') based on the quality of evidence produced, taking into consideration the research design and rigor. The highest rank ('1-') was achieved by randomised controlled trials and the lowest ('3') by non-analytical studies or case reports. A subset of selected articles was graded separately by the two authors and the final agreement level was 90%.

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3 | RESULTS

3.1 | General overview

A total of 41 articles met the inclusion criteria and were included in this review. They spanned over an 18-year period (see Figure 2), with a trend towards an increase in the number of publications over time, as evidenced by 66% being published in the last 6 years. The majority of the studies had a cross-sectional study design (66%) and employed a survey-based approach (51%). A total of 20 studies (49%) used an original or modified existing theoretical adoption models.

Studies were published in various journals, with around 32% appearing in medical informatics journals. They focused on different technologies that they labelled and presented using a variety of terms, including ICT, telecare, telemonitoring, mobile health, and telehealth (referred to as general telehealth, home telehealth services or telemedical) (Figure 3). Other general technologies (e.g. wireless sensor networks, online health management tools, general E-health and mobile remote presence) were grouped under an 'Other' category; two papers did not specify the technology being investigated. The majority of the studies were conducted in Europe and the US (Figure 4).

Table 1 presents a general overview of the studies included in this review (population/specific health condition, study design/level of evidence, sample size, mean age, technology and study duration). While around half the selected studies had a post-assessment design, the remaining employed pre/post-assessment or pre-assessment only. The majority (68%) used a quantitative study design. Overall, the quality assessment of the articles published in this area showed a low to moderate level of evidence (Table 1). The sample sizes also varied considerably, with an average of 195 participants and a mean age of 71.9 years (in studies that reported age). The duration of the studies ranged approximately from 1 month to 4 years; many articles did not indicate the duration of their studies clearly.

3.2 | Technology types and adoption models/frameworks

Multiple studies used and tested existing models, several of which were adapted from the information systems literature. The Technology Adoption Model/Technology Acceptance Model (TAM) was used for analysing technology adoption in several articles. It is grounded on two fundamental constructs: perceived usefulness, which refers to the individual perceiving that the system/technology is useful to them and improves their performance; and perceived ease-of-use, which refers to a person believing that the system/technology does not require a lot of effort and is easy to use (Davis, 1989).

The Unified Theory of Acceptance and Use of Technology (UTAUT) was introduced in 2003 (Venkatesh, Morris, Davis, & Davis, 2003), in an attempt to combine the contributions of different adoption models in the information systems literature. It includes four constructs hypothesised to be predictors of behavioural intention to use a technology: (a) performance expectancy, which is an individuals' belief of the extent to which the system helps in attaining performance benefits (similar to perceived usefulness in TAM); (b) effort expectancy, which is the extent of ease-of-use of the system (similar to perceived ease-of-use in TAM); (c) social influence, which is an individual's perception of whether people important to him/her think that the system should be used; and (d) facilitating conditions, which is an individual's belief of whether conditions exist to support system use. In 2012, and in the light of the proliferation of technologies outside the boundaries of organisations, the Unified Theory of Acceptance and Use of Technology 2 (UTAUT2) model—an extension of UTAUT—was proposed in the context of consumer applications (Venkatesh, Thong, & Xu, 2012). It focuses on the adoption factors on a user-level outside an organisational context (i.e. consumers) and incorporates, in addition to the original UTAUT constructs, three additional ones: hedonic motivation, price/cost value and habit (Venkatesh et al., 2012). Since some studies included in this SR used a combination of UTAUT and

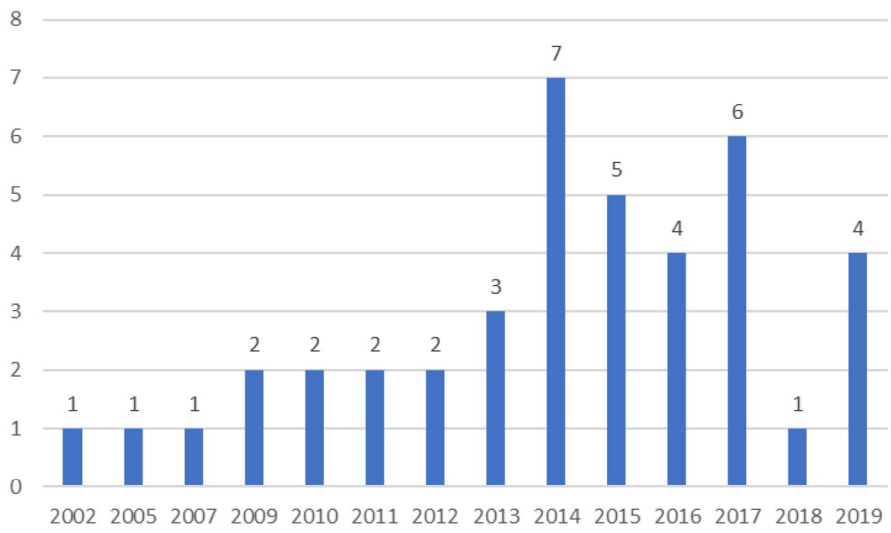

FIGURE 2 Number of studies investigating HIT adoption by seniors over time [Colour figure can be viewed at wileyonlinelibrary.com]

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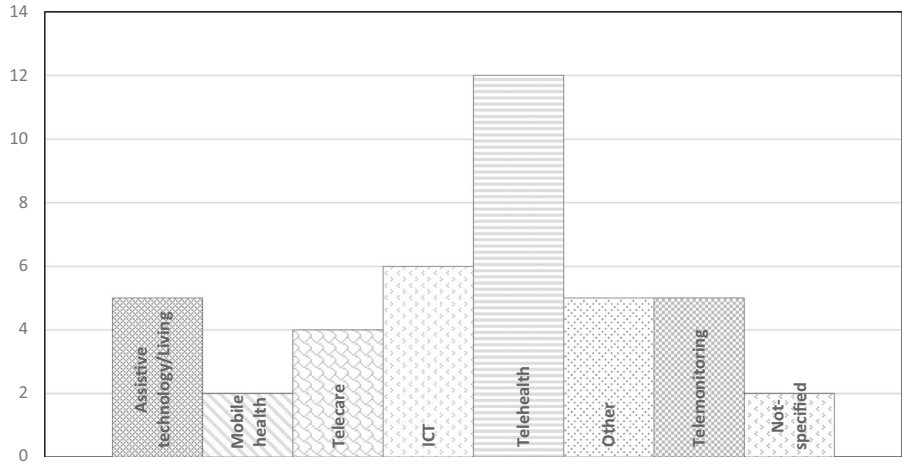

FIGURE 3 Number of studies investigating different types of HIT used by seniors in the community (as defined in the original studies)

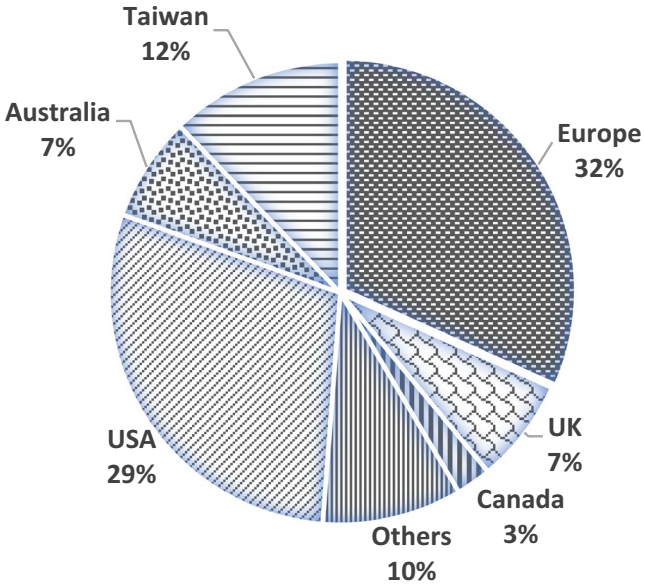

FIGURE 4 Distribution of studies on HIT adoption and seniors by countries of origin [Colour figure can be viewed at wileyonlinelibrary.com]

UTAUT2, we will be referring to these two models in this paper as UTAUT(2).

Table 2 presents an overview of the technologies and adoption models investigated in the empirical studies included in this SR. Given the variety of terminology used by the authors of the original studies in relation to the technology names/labels, and in order to simplify this information, we combined the terms referring to the same technology under one of the respective eight categories of HIT presented in Table 2. The UTAUT(2) model and the TAM were the most frequently used models across different technologies. The Health Information Technology Acceptance Model (HITAM) and Theory of Planned Behavior (TPB) were used specifically in studies investigating telemonitoring and ICT adoption respectively. Other models (e.g. IS success model, Diffusion of Innovation (DoI)) were each used in only one study, and the Cycle of Technology Acquirement by Independent-Living Seniors(C-TAILS) model was the only model developed specifically for a study on technology acquirement by seniors. Interestingly, 21 out of 41 studies included in this review did not consider or use any HIT adoption model.

3.3 | Major factors affecting the adoption of HIT by seniors in the community

Table 3 presents details on the adoption factors/constructs examined in the studies included in this review and provides an overview of those that reported significant results based on the various models tested and technologies investigated. The two constructs that were mostly examined in the literature were performance expectancy and effort expectancy, with a considerable number of studies reporting significant positive impacts of the former and significant negative impacts of the latter on the adoption of technologies by seniors.

3.3.1 | Performance expectancy/ perceived usefulness

Performance expectancy (also referred to as perceived usefulness) was included in a variety of adoption models (UTAUT(2), TAM, C-TAILS, HITAM) investigating the adoption of different types of technologies (e.g. ICT, telemonitoring) by seniors. This construct was assessed in relation to service usefulness in daily life, increase in productivity/quality of life, and quicker and more convenient accomplishment of seniors' daily requirements (Cimperman, Brenčič, & Trkman, 2016; Hoque & Sorwar, 2017; Kohnke, Cole, & Bush, 2014). Performance expectancy was found to be positively related to HIT adoption by elderly people in 29 studies (71% of total articles), 16 of which reported significant positive impacts on seniors' adoption of the respective technologies

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TABLE 1 Overview of studies on HIT adoption among seniors in the community

Reference	Population/specific health condition	Study design/level of evidence	Sample size	Mean age	Technology	Study duration
Alsulami and Atkins (2016)	Seniors	Post-assessment (Quantitative) 2+	194	N/A[a]	Ambient assisted living	N /A[a]
Beer and Takayama (2011)	Seniors	Post-assessment (Qualitative) 2–	12	73.4	Mobile remote presence	N /A[a]
Bertera et al. (2007)	Seniors with chronic disease	Pre-assessment (Quantitative) 2+	85	73	Telecare/environmental sensors	4 months
Blanson Henkemans et al. (2014)	Seniors	Pre-assessment (Quantitative) 2+	28	67	Adaptive Computer Assistant/ICT	N/A[a]
Cabrita et al. (2019)	Seniors with chronic disease	Pre/Post-assessment (Qualitative) 2–	12	69	Ambulatory/ Mobile technology	1 month
Cimperman et al. (2016)	Seniors	Pre-assessment (Quantitative) 2++	400	61.1	Home telehealth services	N/A[a]
Cimperman et al. (2013)	Seniors	Pre-assessment (Qualitative) 2–	87	65	Home telehealth services	N/A[a]
Charness et al. (2016)	Seniors	Post-assessment (Quantitative) 2++	215	53.4	Home healthcare technology	6 months
Choi et al. (2014)	Seniors with depression	Pre/Post-assessment (Quantitative) 1-	121	65.2	Telehealth	24 weeks
Claes et al. (2015)	Seniors	Pre-assessment (Quantitative) 2+	245	72.4	Contactless monitoring sensor	3 months
Czaja et al. (2014)	Seniors with hypertension	Post-assessment (Quantitative) 2–	34	72	Home telehealth	6 months
de Veer et al. (2015)	Seniors	Pre-assessment (Quantitative) 2++	1,014	67	E-health	N/A[a]
Etemad-Sajadi et al (2019)	Seniors	Post-assessment (Quantitative) 2+	213	82.1	Assistive alarm/ telecare/sensors	N/A[a]
Hamblin (2017)	Seniors susceptible to falls or with cognitive impairment	Pre/Post-assessment (Qualitative) 2–	60	N/A[a]	Telecare	9 months
Heart and Kalderon (2013)	Seniors	Post-assessment (Quantitative) 2+	123	80	ICT	2 years
Heerink et al. (2010)	Seniors	Pre/Post-assessment (Quantitative) 2++	188	N/A[a]	Assistive social agent technology	N/A[a]
Hoque and Sorwar (2017)	Seniors	Pre-assessment (Quantitative) 2+	274	N/A[a]	Mobile health	N/A[a]
Hsieh et al. (2015)	Seniors	Post-assessment (Quantitative) 2+	336	N/A[a]	Telehealth	N/A[a]
Jaana et al. (2017)	Seniors with chronic diseases	Post-assessment (Quantitative) 2+	23	75.2	Telemonitoring	6 months
Keränen et al. (2017)	Seniors with and without frailty	Post-assessment (Quantitative) 2+	794	81.5	ICT	3 months
Kohnke et al. (2014)	Seniors	Post-assessment (Quantitative) 2–	126	N/A[a]	Telehealth/ Telemedicine	3 months
Loh et al. (2009)	Seniors	Post-assessment (Mix) 2+	53 + 6 focus group	N/A[a]	ICT	N/A[a]
Lu et al. (2014)	Seniors with chronic diseases	Post-assessment (Qualitative) 3	20	67	Home telehealth	3 months
Mann et al. (2002)	Seniors with chronic diseases	Pre-assessment (Quantitative) 2–	71	73.5	Health monitoring devices	N/A[a]
McCreadie and Tinker (2005)	Seniors	Post-assessment (Qualitative) 2–	67	83.5	Assistive technology	N/A[a]
Middlemass et al. (2017)	Seniors with long-term condition	Post-assessment (Qualitative) 1-	53	79.5	Telemonitoring	4 years
Mitzner et al. (2010)	Seniors	Post-assessment (Qualitative) 2++	113	73	N/A[a]	N/A[a]
Peek et al. (2017)	Seniors	Pre/post-assessment (Qualitative) 1-	33	76.1	N/A[a]	3 years
Peeters et al. (2012)	Seniors with chronic diseases	Post-assessment (Quantitative) 2+	254	77.8	Home telecare	N/A[a]
Russell et al. (2015)	Seniors	Pre-assessment (Quantitative) 2+	306	59	Home telehealth	N/A[a]

(Continues)

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TABLE 1 (Continued)

Reference	Population/specific health condition	Study design/level of evidence	Sample size	Mean age	Technology	Study duration
Steele et al. (2009)	Seniors	Pre-assessment (Qualitative) 3	13	N/A[a]	Wireless sensor networks	N/A[a]
Tsai (2014)	Seniors with chronic diseases	Post-assessment (Quantitative) 2+	365	N/A[a]	Telehealth	N/A[a]
Tseng et al. (2013)	Seniors	Post-assessment (Quantitative) 2–	32	N/A[a]	Health monitoring system	N/A[a]
van Hoof et al. (2011)	Seniors	Pre/Post-assessment (Qualitative) 2–	18	N/A[a]	Ambient intelligence technology	23 months
Van Houwelingen et al. (2018)	Seniors	Pre/Post-assessment (Mix) 1-	256 & 15	70 & 73.2	Telehealth	N/A[a]
Vroman et al. (2015)	Seniors	Post-assessment (Quantitative) 2+	198	76	ICT	N/A[a]
Wong et al. (2012)	Seniors	Post-assessment (Quantitative) 2++	121	70.5	Telehealth/Telemedical	1 year
Zacharopoulou et al. (2019)	Seniors with dementia	Post-assessment (Quantitative) 2+	779	77.7	ICT	15 Month
Zettel-Watson and Tsukerman (2016)	Seniors with chronic diseases	Post-assessment (Quantitative) 2–	169	70.8	Online health management tools	N/A[a]
Zhang et al. (2014)	Seniors with dementia	Pre-assessment (Quantitative) 2–	40	N/A[a]	Assistive technology	5 weeks
Zhou et al. (2019)	Seniors with chronic diseases	Pre-assessment (Quantitative) 2+	436	N/A[a]	Telehealth	N/A

[a]N/A, Not Applicable.

investigated (e.g. de Veer et al., 2015; Heart & Kalderon, 2013; Hoque & Sorwar, 2017). For instance, in a survey of 274 seniors (60 years and older), Hoque and Sorwar (2017) demonstrated that performance expectancy had a significantly positive impact on seniors' behavioural intention to adopt Mobile Health (M-Health). Heart and Kalderon (2013) also reported that ICTs were adopted by seniors who were strongly motivated by performance expectancy (Heart & Kalderon, 2013). Van Houwelingen, Ettema, Antonietti, and Kort (2018) used a mixed-method approach and also reported that performance expectancy affects seniors' intention to use telehealth/video conferencing (van Houwelingen et al., 2018).

3.3.2 | Effort expectancy/perceived ease-of-use

Effort expectancy (also referred to as perceived ease-of-use)was found to have a negative impact on the adoption of HIT by elderly people in 29 studies (71%), of which 14 studies reported significant negative effects on the adoption of various types of technology (e.g. ICT, E-health, telemonitoring). Effort expectancy was assessed in a variety of approaches including how simple the technology is to use, learning and convenience in using it and the simplicity of operating it (e.g. Cimperman et al., 2016; de Veer et al., 2015; Kohnke et al., 2014; Tseng, Hsu, & Chuang, 2013). For example, de Veer et al. (2015) found that performance expectancy and effort expectancy are highly related to the intention to use E-Health technologies, with around 63% of participants indicating they would 'definitely or probably use' this technology if offered (de Veer et al., 2015). Tsai (2014) used a self-administered questionnaire among 365 participants in central Taiwan and also found that perceived ease of use and perceived usefulness influence the intention to use telehealth technology (Tsai, 2014).

3.3.3 | Social influence

Social influence was investigated in 15 studies that considered various models (i.e. C-TAILS, UTAUT(2), TAM and TPB) to explain the adoption of different technologies (e.g. ICT, telemonitoring, E-health, M-health) by seniors. It was generally assessed in relation to the influence people who are important to seniors (e.g. family members or friends) have on their adoption decisions (Hoque & Sorwar, 2017; Kohnke et al., 2014; Tseng et al., 2013). Despite the generally positive reported impacts of social influence on users' behavioural intention to adopt HIT, only three studies found significant positive effects (Hoque & Sorwar, 2017; Kohnke et al., 2014; Tseng et al., 2013), and two studies with a higher level of evidence (Cimperman et al., 2016; de Veer et al., 2015) reported no effect on HIT acceptance behaviour. For instance, de Veer et al. (2015) found no causal relationship between social influence and the intention to use E-health technologies among persons 57–77 years old (de Veer et al., 2015). Another study by Cimperman et al.(2016) involving 400 participants, aged 50 years and older, found that social influence was not related to seniors' acceptance

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behaviour, although doctors' recommendations strongly impacted performance expectancy (Cimperman et al., 2016). Interestingly, Zettel-Watson and Tsukerman (2016) also reported that doctors' recommendations have an important effect in the early stages of online management tools use; this effect, however, appeared to decrease over time (Zettel-Watson & Tsukerman, 2016).

3.3.4 | Facilitating conditions

Facilitating conditions were examined in the UTAUT(2) and TAM models across studies investigating different technologies (telemonitoring, E-Health). One study used the necessary mental ability, knowledge-to-use technology, available support and assistance in difficulties as indicators of facilitating conditions (Kohnke et al., 2014). Another study by Jaana et al. (2017) assessed facilitating conditions in relation to the available technical support and training that elderly people received when using telemonitoring (Jaana et al., 2017). Overall, facilitating conditions were reported to have a positive impact on seniors' behavioural intention to use HIT in eight studies, with three demonstrating significant positive effects. However, the impact of facilitating conditions was less commonly reported in comparison to performance expectancy, effort expectancy and social influence, although significant in a few studies. Cimperman et al. (2016) reported that facilitating conditions (e.g. technical support) had a 'direct impact' on the acceptance behaviour of older users of home telehealth services; they improved seniors' behavioural intention to use this technology (Cimperman et al., 2016). Kohnke et al. (2014) also indicated that facilitating conditions along with performance expectancy, effort expectancy and social influence as predictors for seniors' acceptance of telemedicine equipment, with a significant impact on behavioural intention to use them (Kohnke et al., 2014). Only one study reported that facilitating conditions do not have a significant impact on seniors' behavioural intention to adopt M-Health technology (Hoque & Sorwar, 2017).

3.3.5 | Price/cost value

The price/cost value was referred to studies that considered almost all types of technologies and appeared in various adoption models including UTAUT(2), C-TAILS and DoI. The cost of HIT relates to the installation, service, repairs and maintenance of the technology (Alsulami & Atkins, 2016; Lu, Chi, & Chen, 2014; Steele, Lo, Secombe, & Wong, 2009). Overall, 15 studies reported a negative impact of price/cost value on HIT adoption by seniors, of which six studies indicated significant negative effects. Alsulami and Atkins (2016) found that the cost of assistive technology creates a big challenge for seniors' behavioural intention to use it, and argued that such technology should be affordable or free to use (Alsulami & Atkins, 2016). Likewise, Lu et al. (2014) described the cost as the most important factor affecting the intention to use home telehealth care among chronically ill seniors; they indicated concerns over reimbursement policy changes that

TABLE 2 Distribution of studies that focused on different HIT and adoption models

HIT types	Adoption model									Number of studies that referred to the technology
	C-TAILS	HITAM	UTAUT(2)	TAM	TPB	Diffusion of innovation	Obtrusiveness model	IS success model	No model	
ICT			✓		✓				✓ (4)[a]	6
Telemonitoring		✓	✓						✓ (3)[a]	5
Telecare				✓		✓	✓		✓	4
Assistive technology/living			✓						✓ (4)[a]	5
Mobile health			✓						✓	2
Telehealth			✓ (3)[a]	✓ (4)[a]				✓	✓ (4)[a]	12
Other			✓ (2)[a]						✓ (3)[a]	5
Not-specified	✓								✓	2
Total Number of Studies	1	1	9	5	1	1	1	1	21	41
Number of studies referring to adoption models: Developed (D), Used (U), Modified (M)	D:1	U:1	U:3, M:6	M:5	U:1	U:1	M:1	U:1	0	D:1 U:7 M:12

[a]The number of studies (when >1) are included in the brackets.

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TABLE 3 Adoption factors and constructs investigated in empirical studies on HIT adoption by seniors

Factors &constructs reported in empirical studies	TP[a]	SP[b]	TN[c]	SN[d]	NN[e]	Adoption models considered in the studies	Technology type
Performance expectancy/ perceived usefulness	29	16	–	–	–	C-TAILS, HITAM, UTAUT(2), TAM, No-model, IS success model, TPB, DoI	ICT, telemonitoring, E-health, M-health, telehealth, telecare, assistive technology
Effort expectancy/perceived ease-of-use	–	–	29	14	–	C-TAILS, HITAM, UTAUT(2), TAM, No-model, IS success model, TPB, DoI, Obtrusiveness model	ICT, telemonitoring, E-health, M-health, telehealth, telecare, assistive technology
Social influence	15	3	–	–	2	C-TAILS, UTAUT(2), TAM, No-model, TPB	ICT, telemonitoring, E-health, M-health, home telehealth, telecare, assistive technology
Facilitating conditions	8	3	–	–	1	UTAUT(2), TAM, No-model	ICT, telemonitoring, E-health, M-health, home telehealth, telecare, assistive technology
High price/Cost value	–	–	15	6	1	C-TAILS, UTAUT(2), DoI, Obtrusiveness model, No-model	ICT, telemonitoring, E-health, M-health, home telehealth, telecare, assistive technology
Privacy/Security	2	–	9	4	3	UTAUT(2), TAM, TPB, Obtrusiveness model, DoI, No-model	ICT, telemonitoring, telecare, E-health, assistive technology, M-health, home telehealth
Habit	–	–	1	–	1	UTAUT(2)	Health monitoring
Intrinsic technology quality	6	2	–	–	–	TAM, IS success model, No-model, DoI, TPB, UTAUT(2)	ICT, telemonitoring, telecare, assistive technology, M-health, telehealth
Self-efficacy	8	3	–	–	1	HITAM, UTAUT(2), TAM, No-model	ICT, telemonitoring, E-health, M-health, telehealth, telecare, assistive technology
Seniors-centred product design	12	–	–	–	–	C-TAILS, UTAUT(2), TAM, Obtrusiveness model, No-model	ICT, telemonitoring, telecare, E-health, M-health, telehealth, assistive technology
Experience/Training	6	–	–	–	2	UTAUT(2), TAM, TPB, No-model	ICT, telecare, E-health, telehealth, assistive technology
Age	4	1	3	1	4	UTAUT(2), TAM, TPB	ICT, telemonitoring, telecare, E-health, M-health, telehealth
Higher education	5	1	–	–	7	UTAUT(2), TAM, TPB	ICT, telemonitoring, telecare, E-health, M-health, telehealth
Gender	2	–	–	–	9	TAM, UTAUT(2), No-model	ICT, telemonitoring, telecare, M-health, telehealth
Health conditions	2	–	1	1	4	UTAUT(2), No-model	ICT, telemonitoring, telecare, assistive technology, M-health, telehealth
Technology anxiety	–	–	7	1	2	UTAUT(2), TAM, No-model	ICT, telemonitoring, telecare, assistive technology, M-health, telehealth

[a]TP: Total # Studies Showing *Positive Effects*.
[b]SP: Total # Studies Showing *Significant Positive Effects (a subset of TP)*.
[c]TN: Total# Studies Showing *Negative Effects*.
[d]SN: Total # Studies Showing *Significant Negative Effects (a subset of TN)*.
[e]NN: Total # Studies Showing *no evidence of Effects*.

preclude coverage of HIT as it was in pilot testing (Lu et al., 2014). In addition, Claes et al. (2015) discussed the potential of low technology cost, the usefulness and availability of collected information, and functional requirements in improving seniors' perceptions and attitudes toward contactless sensors (Claes et al., 2015). Their survey of 245 older adults (≥60 years) found that almost 70% considered price/cost value in their decision to accept this technology, and indicated unwillingness to bear the costs of the system's maintenance or to ask for co-financing or family assistance (Claes et al., 2015). Hence, the evidence shows a negative effect of higher costs on HIT adoption among seniors.

3.3.6 | Privacy/security

Last, based on the studies included in this review, the evidence on the impacts of privacy/security concerns on HIT adoption by

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seniors remains inconclusive. Privacy/security concerns were assessed in a variety of approaches, including seniors' need for privacy in sharing health information with others through ICT (Cimperman et al., 2016), and worries about data security and home security in using ambient assisted living technology (Alsulami & Atkins, 2016). Concerns over privacy/security were found to be barriers to HIT adoption in nine studies (Alsulami & Atkins, 2016; Beer & Takayama, 2011; Cimperman et al., 2016; Cimperman, Brenčič, Trkman, & Stanonik, 2013; Claes et al., 2015; Loh, Flicker, & Horner, 2009; Mitzner et al., 2010; van Houwelingen et al., 2018; Zettel-Watson & Tsukerman, 2016), out of which four studies reported significant negative effects. Specifically, 15% of participants in one study reported privacy concerns over accepting mobile remote presence systems (Beer & Takayama, 2011). In another study, perceived privacy concerns had a negative impact on seniors' desire to share health information with family members (Cimperman et al., 2016), and a third study found that safety/security represent important concerns limiting seniors' adoption of assistive technology (Alsulami & Atkins, 2016). Nevertheless, two other studies postulated that HIT might, in fact, increase privacy/security and improve seniors' attitudes towards HIT adoption (Lu et al., 2014; van Hoof, Kort, Rutten, & Duijnstee, 2011); these studies, however, had a relatively low level of evidence. For instance, Lu et al. (2014) indicated that home telehealth might enhance the sense of support and security (i.e. safety from negative health consequences) and improve the attitude of seniors with chronic conditions towards HIT adoption; this is particularly relevant given the sense of support that patients feel when their health condition is being monitored continuously (Lu et al., 2014). Moreover, van Hoof et al. (2011) found that privacy is not perceived as a barrier, explaining that the sense of safety and security for older users may be improved with the use of ambient intelligence technology and that privacy concerns did not present an issue for actual users (van Hoof et al., 2011). On the other hand, three studies found that privacy had no effect on seniors' HIT adoption, although they emphasised that this concern should not be overlooked (Charness, Best, & Evans, 2016; Hamblin, 2017; Steele et al., 2009). Therefore, given the variation in the levels of evidence in the empirical studies and the divergence in the direction of the results, the impacts of privacy/security in relation to HIT adoption among seniors remain inconclusive.

3.4 | Factors presenting both direct and moderating effects on HIT adoption by seniors

A moderating variable typically affects the relationship between an independent and a dependent variable, by reducing or strengthening this relationship (MacKinnon, 2011). A number of moderator factors/variables were reported in the literature when examining the relationship between the adoption factors presented above and seniors' behavioural intention to use, or actual use of the various technologies. These factors/variables presented in this section were reported as having both a direct effect on the behavioural intention to use the technology and a moderating influence that increases/decreases the impacts of other adoption factors on technology use.

3.4.1 | Habit

Habit, also described as individuals acting on the basis of prior behaviour–future behaviour relationship (Kim & Malhotra, 2005), was included in the UTAUT(2) adoption model tested in the literature in the context of seniors. Russell et al. (2015) found that traditional healthcare habits, dissatisfaction with current service and online behaviour predict telehealth adoption (Russell et al., 2015). Nevertheless, the number of studies reporting impacts of habit on HIT adoption was small, and their level of evidence generally low; hence, the evidence in this area remains inconclusive.

3.4.2 | Intrinsic qualities of the technology

Six studies examined the intrinsic qualities of the technology in relation to its adoption by elderly people. The intrinsic quality of the technology was investigated in relation to system quality, information quality and service quality (e.g. Hsieh, Tsai, Chih, & Lin, 2015), as well as the quality of healthcare (e.g. Peeters, de Veer, van der Hoek, & Francke, 2012). Only one study, involving 336 participants in a rural community in Taiwan, found that among three intrinsic qualities considered (i.e. system, information and service quality), system quality had the strongest overall positive effect on the intention to use telehealth (Hsieh et al., 2015). Zhou et al. (2019) applied an extended TAM model and reported that service satisfaction, effort expectancy and information quality had significant impacts on telehealth adoption by elderly patients with chronic conditions (Zhou et al., 2019). No similar evidence, however, was reported in other studies.

3.4.3 | Self-efficacy

Self-efficacy refers to a person's capacity to perform a specific behaviour or his/her ability to achieve objectives (Compeau & Higgins, 1995). Self-efficacy was investigated based on task completion by seniors without assistance or through using available resources, such as instructions or others' help (Kohnke et al., 2014). Eight studies reported positive direct or moderating effects of self-efficacy on behavioural intention to use HIT among elderly people (de Veer et al., 2015; Kohnke et al., 2014; Lu et al., 2014; Middlemass, Vos, & Siriwardena, 2017; Mitzner et al., 2010; Steele et al., 2009; Tsai, 2014; van Houwelingen et al., 2018). For instance, de Veer et al. (2015) found that self-efficacy was related to the intention to use E-Health technology by community-dwelling older people (de Veer et al., 2015). Self-efficacy was also reported in two other studies as having a strong moderating effect on behavioural intention

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to use telemedicine and telehealth (Kohnke et al., 2014; Tsai, 2014). Tsai (2014) indicated that self-efficacy has a significant impact on the perceived ease of use (Tsai, 2014), and Kohnke et al. (2014) reported a strong relationship between the constructs examined in the UTAUT model, self-efficacy and behavioural intention to use the telemedicine equipment (Kohnke et al., 2014). Only one study found no effects of self-efficacy on HIT adoption (Cimperman et al., 2013).

3.4.4 | Product design

Product design, which refers to designing products or services according to seniors' needs and capabilities, was included in a variety of models (e.g. C-TAILS, UTAUT(2), and TAM) used in studies involving different types of technology (e.g. ICT, telemonitoring, telecare). Product design was assessed in relation to adaptability with existing systems (Blanson Henkemans et al., 2014) and user-centred requirement design (Czaja, Lee, Arana, Nair, & Sharit, 2014). It was reported as a moderating variable in 12 studies that found an impact of customised design on seniors' HIT adoption factors; however, none of these studies reported significant effects (Blanson Henkemans et al., 2014; Cabrita, Tabak, & Vollenbroek-Hutten, 2019; Claes et al., 2015; Czaja et al., 2014; Loh et al., 2009; Lu et al., 2014; Mann, Marchant, Tomita, Fraas, & Stanton, 2002; McCreadie & Tinker, 2005; Middlemass et al., 2017; Steele et al., 2009; Tseng et al., 2013; van Hoof et al., 2011). For example, Steele et al. (2009) found that design simplicity had a positive effect on the relationship between effort expectancy and behavioural intention to use, and indicated that investigating users' needs and requirements are vital for a variety of health technologies such as wireless sensor networks (Steele et al., 2009). This is similar to what was reported by Tseng et al. (2013), who found that a user-friendly design of intelligent health monitoring systems had an impact on the acceptance of these systems by elderly people (Tseng et al., 2013). Likewise, Czaja et al. (2014) discussed the importance of incorporating a user-centred design for seniors in the manufacturing of home telehealth systems, in order to accommodate their diverse needs (Czaja et al., 2014).

3.4.5 | Experience and training

Experience and training also play an important role in HIT adoption by seniors; these were assessed in relation to seniors' previous experience with similar ICT (Czaja et al., 2014; Heart & Kalderon, 2013), the adequate training in relation to ICT use or otherwise previous inadequate training (Loh et al., 2009). Six studies reported a positive effect of seniors' experience and training on their adoption of HIT, but there were no studies showing significant findings at this level (Alsulami & Atkins, 2016; Czaja et al., 2014; de Veer et al., 2015; Heart & Kalderon, 2013; Loh et al., 2009; van Houwelingen et al., 2018). For instance, one study found that training and ease of experience improve users' HIT perception of ambient assisted living and online health management (Alsulami & Atkins, 2016).

3.5 | Factors presenting only moderating effects on HIT adoption by seniors

In addition to habit, intrinsic technology quality, self-efficacy, senior-centred product design and experience/training, previously discussed as having both direct and moderating effects, we present in the following sections the factors/variables that were found to only have moderating effects on seniors' behavioural intention to use or actual use of various HIT.

3.5.1 | Socio-demographic variables (age, higher education, gender)

Based on the existing literature, there is inconsistent and weak evidence on the impacts of socio-demographic variables (age, higher education, gender) and health conditions on HIT adoption, with both positive and negative effects reported in the empirical studies examined in this review (de Veer et al., 2015; Wong et al., 2012; Zacharopoulou et al., 2019). The findings in relation to the impacts of socio-demographic characteristics on the relationship between adoption factors and use of HIT varied. Heart and Kalderon (2013) reported that older people showed a lower intention to use health-related ICTs (Heart & Kalderon, 2013), while Keränen et al. (2017) found that the use of ICT by seniors with and without frailty varied widely and age, education and health condition only partly explained this variation (Keränen et al., 2017). Etemad-Sajadi and Dos Santos (2019) used the modified TAM and reported that gender had a significant impact on the 'perceived intrusiveness', with men considering HIT innovations (e.g. assistive alarm, telecare and sensors) more obtrusive than women. In addition, seniors who live alone tend to be more likely to accept HIT than those living with a partner (Etemad-Sajadi & Dos Santos, 2019). On the other hand, the study by Bertera, Tran, Wuertz, and Bonner (2007) showed no significant impacts of these characteristics on HIT use by seniors (Bertera et al., 2007). In addition, Blanson Henkemans et al. (2014) reported that personal characteristics (i.e. age, gender, health condition) can influence users' success with ICT (Blanson Henkemans et al., 2014) and de Veer et al.(2015) found that men tend to use E-health more than women and discussed the importance of focusing on gender and educational level when working towards improving HIT acceptance (de Veer et al., 2015). Along the same line, two other studies reported that higher education in general increased the use of ICTs (Vroman, Arthanat, & Lysack, 2015; Zacharopoulou et al., 2019). In summary, the evidence on the moderating effect of socio-demographic characteristics is inconsistent and remains unclear in relation to HIT adoption and seniors.

3.5.2 | Technology anxiety

Last, technology anxiety was assessed in the original studies included in this review as seniors feeling 'nervous', 'worried', 'uncomfortable', 'uneasy' or 'confused' (e.g. Hoque & Sorwar, 2017; Kohnke et al., 2014; Russell et al., 2015). Technology anxiety was found to have a negative

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moderating effect on effort expectancy and behavioural willingness to use technology; only one study, however, reported significant effects (Hoque & Sorwar, 2017). For instance, Steele et al. (2009) discussed participants' concerns about technology anxiety in relation to a wireless sensor network. Steele et al. (2009) and Cimperman et al. (2016) reported that computer anxiety negatively moderated the impact of effort expectancy for seniors using home telehealth services (Cimperman et al., 2016). Nevertheless, the number of studies reporting negative impacts of technology anxiety on HIT adoption was low, and two studies (Heerink, Kröse, Evers, & Wielinga, 2010; Russell et al., 2015) indicated no evidence at this level.

3.6 | Summary of evidence: a guide for future research

In an attempt to summarise the main findings in this SR and guide future research on HIT adoption and seniors, we present a synthesis/synopsis of the evidence in this area (Table 4). To simplify the process, we proposed a categorisation of the level of evidence (high/moderate/low), taking into consideration the number of studies that examined each construct/factor, the rigor of the respective studies and the consistency of the findings (Table 4). In addition to the constructs/factors adapted from the information systems literature and applied in the context of healthcare to better explain HIT adoption by seniors (e.g. performance expectancy, effort expectancy, social influence, facilitating conditions, experience/training, self-efficacy, technology anxiety, habit, intrinsic technology quality and price/cost value), this review highlights the importance of considering privacy/security and product design when evaluating HIT adoption by seniors. This summary table constitutes a unifying starting point for researchers interested in investigating HIT adoption by seniors and can guide future studies in this area.

4 | DISCUSSION

HIT represent a broad range of technologies that aim to support the care and management of various health conditions among elderly people. To date, the results from empirical studies that investigated the factors affecting the adoption of HIT by seniors have been scattered, with limited knowledge about the level of evidence in the existing literature. No prior review has specifically examined the different adoption models studied in the literature in relation to HIT that support seniors, or synthesised the evidence on the significant factors affecting the adoption of these technologies. This SR contributes to this area and critically appraises and synthesises existing evidence on HIT adoption by seniors, which may inform future research and guide policies and efforts targeting this group of users.

This SR did not identify particular patterns, or significant differences, in relation to the factors that affect the adoption of various types of HIT, or different age groups of seniors. The findings of this SR reveal that around 51% of the studies on seniors' HIT adoption did not consider a technology adoption model/framework. Among the studies that otherwise used a conceptual model/framework, only one was developed specifically for the context of seniors (Peek et al., 2017). The choice of the adoption model considered in the empirical studies was often justified by the authors in the context of their respective research. However, some technologies (e.g. telehealth) and IT adoption models (e.g. UTAUT(2)) were more frequently studied than others. It is also important to note the significant variation in the terminology and extent of description of the technologies investigated in the original articles, which is an indication of fragmentation and inconsistency in the field; this makes comparisons across studies and synthesis of findings very challenging.

In order to address the research questions in this SR, we synthesised the empirical evidence that investigated 16 adoption constructs/factors and moderators considered in the literature. Our findings show that *performance expectancy* (perceived usefulness), *effort expectancy* (perceived ease of use) and *price/cost value* were the most investigated, with evidence of their significant impacts on HIT adoption. S*ocial influence* and *product design* were also commonly studied, but with a lower number of studies reporting significant effects of these factors. Other factors including *facilitating conditions*, *self-efficacy*, *intrinsic qualities*, *experience/training* and *technology anxiety* were also considered, but the evidence on their impacts remains either weak or limited. Interestingly, there was no consistent nor strong evidence on the effects of socio-demographic variables (e.g. *age*, *gender*, *higher education)*, *health condition*, *habit* and *privacy/security concerns* on HIT adoption among seniors. Given the importance and relevance of frailty among seniors, its relation to physical capacity and the ability to live at home, and the potential to reverse it if detected early (e.g. Kojima, 2018; Sørbye, Sverdrup, & Pay, 2018; Xue, 2011), we propose that future researches examine the impacts of frailty on HIT adoption.

Despite increased efforts in recent years to investigate the adoption of HIT by seniors, evidence in this area remains scattered with a constant focus on using existing models and common rationalistic approaches in investigating this topic. To date, we seem to be unable to explain the low level of HIT adoption by seniors and the variation in technology use among this group, which is critical for fully leveraging the potential of these tools to better support elderly care in the community. Although existing IT adoption models have made contributions to better understand HIT adoption by seniors, we have reached a point of minimal advancement in relation to our ability to explain the relatively low level of HIT adoption by this particular group of users in the context of diverse healthcare systems. This is further emphasised by inconsistent and inconclusive evidence in the literature in relation to the majority of the factors that have been considered. The question that presents itself is whether research in this area, within the existing paradigm, is sufficient to reach a better understanding of seniors' behaviour vis-à-vis HIT, and adequately address their care needs using these tools. We posit that research on seniors' adoption of HIT should extend beyond the traditional IT

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TABLE 4 Summary of existing evidence on HIT adoption factors in the literature

	Constructs/factors	Description in empirical studies	Predominantly positive impacts	Predominantly negative impacts	Inconclusive evidence/no impacts
Direct effect	Performance expectancy	Factor	SE		
	Effort expectancy	Factor		SE	
	Social influence	Factor	ME		
	Facilitating conditions	Factor	ME		
	High price/Cost value	Factor		SE	
	Privacy/Security	Factor			X
Direct and moderating effect	Habit	Factor/moderator			X
	Intrinsic technology quality	Factor/moderator	LE		
	Self-efficacy	Moderator/factor	ME		
	Senior-centred product design	Moderator/factor	ME		
	Experience/Training	Moderator/factor	LE		
Moderating effect	Age	Moderator			X
	Higher education	Moderator			X
	Gender	Moderator			X
	Health conditions	Moderator			X
	Technology anxiety	Moderator		LE	

Note: X: Inconclusive Evidence/No Evidence may be attributed to a low number of studies reporting either positive or negative impacts, or a large number of studies reporting no impacts.
SE, ME, LE are based on the number of studies, their rigour and respective level of evidence.
ME, Moderate Evidence; LE, Low Evidence; SE, Strong Evidence.

adoption paradigm, in order to provide a more complete explanation for the low level of HIT adoption among seniors in today's environment, and the variation in elderly HIT adoption across countries and regions within the same country.

IS research on IT innovations in organisations has challenged existing paradigms in the past in an attempt to expand the perspective and investigation lens and contribute to a more comprehensive understanding of the diffusion of innovations in these settings (e.g. Kaganer, Pawlowski, & Wiley-Patton, 2010; Ramiller & Swanson, 2003). In the context of consumer HIT applications and seniors, would a paradigm shift contribute to an improved collective understanding of the role that HIT can play in supporting seniors' care in the community and the conditions that would enable adoption and optimal benefits from these technologies?

Limited work has been done in the past to investigate seniors' HIT adoption, taking into consideration their collective environment, which may be shaped by the web/network of interactions between them and their surroundings. In a complex environment like healthcare in which elderly interact with numerous persons/stakeholders contributing to their care, it is important to study the interaction and information exchange between them, their family/caregivers, their peers, their health care providers and the technology suppliers. We concur with Schulz et al. (2015) that research on technology adoption and elderly should extend beyond describing the facilitators and barriers. Instead, we need to develop a better understanding of the environment surrounding elderly people and their interaction with this environment (Schulz et al., 2015). Thus, we call for future research to explore the nature and weight of the dyads of interactions that is, senior-family/caregiver, senior–senior, senior–provider, senior–vendor, as well as the most effective mechanisms for communication, which may shape seniors' decisions to use or not use a technology, and hence would inform efforts and strategies in this area. We believe that uncovering the role of communication and interaction between elderly people and their environment would support a 'senior-centred' digital health research agenda that contribute to the care of seniors in the community.

It is important to note that, given the scope of this review, we focused on seniors' HIT adoption and limited our results to the constructs/factors investigated in the healthcare literature with seniors as target participants. Future studies should also investigate the perspectives of caregivers in this area, given their close interaction with and knowledge of seniors and their needs. In addition, healthcare systems are very different across disparate regions and various countries. For example, the cost of HIT may not a concern for seniors residing in areas with state/province-provided healthcare, but it can be an enormous barrier in other regions. Hence, system-level variables may play an important role in determining HIT adoption by seniors, especially at the community level, and variations in policies

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and HIT cost coverage should be carefully examined and discussed when conducting studies in this area.

Last, our review was based on published studies in this area of research, thus potentially presenting limitations inherent to the original biases and variation in the rigour of the empirical studies. In addition, although the impact of different adoption factors did not change significantly across technologies, the number of studies varied considerably based on technology types; this may also affect the generalisability of the findings. For example, the number of studies focusing on M-health was very limited as compared to other technologies like telehealth, thus highlighting the need for more research in these areas and better classification of these technologies.

We consulted five major academic databases in this study, using a variety of keyword combinations. However, the articles included in this review were limited to studies published in English, excluding potentially relevant studies that may have been published in other languages. The level of evidence in the original articles was relatively low moderate, which may limit the confidence in the conclusions presented in this review. Hence, we recommend future empirical studies to consider more rigorous designs/approaches and longitudinal evaluations to assess the factors that significantly affect the adoption of different types of HIT by seniors in the community and the differential impacts that may exist among various groups of seniors.

5 | CONCLUSION

Living independently or remaining in the community can be challenging for seniors, especially if they have additional healthcare needs and different levels of frailty. HIT present opportunities to support seniors by enabling independent living, increased safety, better symptom management and enhanced quality of life. Despite these opportunities, the relatively low level of HIT adoption among this group calls for more rigorous research in this area and qualitative studies that allow a better understanding of this population's perspectives and preferences, taking into consideration the collective environment in which they live and their discourse and interaction/communication within this environment. Importantly, future research on HIT should present a clear description of the technology being investigated and its functionality, and work towards a common classification and terminology that would enable a better understanding of each particular technology potential contribution to elderly care.

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CONFLICT OF INTEREST

There is no conflict of interest to declare.

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ENDNOTE

[1] The 'community dwelling' elderly are group of seniors with age (60 years or older) who live independently. This age group may or may not suffer from any disease with or without need for specific health care requirements Steultjens, Dekker, Bouter, Jellema, Bakker, & Van Den Ende, C. H. (2004). Occupational therapy for community dwelling elderly people: a systematic review. *Age and ageing*, 33(5), 453–460).

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APPENDIX 1

Research keywords.

Group 1 (G1): Acceptance OR Usability OR Use OR Adoption OR "behavioral intention"

Group 2 (G2): Elderly OR Seniors OR Aging OR Old

Group 3 (G3): Information and Communication Technology OR ICT OR Telemonitoring OR "E health" OR E-health OR Ehealth OR Telehealth OR Telemedicine OR "Assistive technology" OR "Personal health record" OR telecare OR "home automation systems" OR "Telehome care" OR "E consultation" OR E-consultation OR "smart home"

Group 4 (G4): Home OR "retirement home" OR "Nursing home" OR "Community dwelling"[1] OR "Community residence " OR "Living in the community " OR "aging in place" OR "independent living" OR Community OR "home care"

Query: Group 1 AND Group 2 AND Group 3 AND Group 4

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